PROJECT BASED EXPERIENTIAL LEARNING REPORT

ON

" Tomato and Potato Leaf Disease Detection "

Under the Course

PROJECT BASED EXPERIENTIAL LEARNING

SUBMITTED BY

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2025-26

THIRD Year Computer Science & Engineering (TY CSE)

Division: B , Semester - II

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2025-2026
Program: B. Tech Computer Science & Engineering
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CERTIFICATE

This is to certify that student Aryaa Katikeri, Rutuja Sankpal, PRN.1012412095, 1012412099 is studying in **TY CSE (Division:)** course in SEM II and he/she has successfully completed and submitted the **Project Based Experiential Learning Report under the Course "Project Based Experiential Learning"** , entitled " Tomato and Potato Leaf Disease Detection " .This study is a partial fulfillment of the degree of Bachelor of Technology in Computer Science and Engineering of **DES Pune University, Pune** during the academic year **2025-2026.**

Prof. Hrishikesh Vanjari	Dr. Prachi Joshi
Mentor	HoS

ACKNOWLEDGMENT

I would like to express my sincere gratitude to my project mentor for their valuable guidance, encouragement, and continuous support throughout the development of this project. Their suggestions and insights played an important role in the successful completion of this work.

I would also like to thank the Department of Computer Engineering of our institution for providing the necessary facilities and resources required for completing this project.

I am grateful to all the faculty members who provided valuable suggestions and guidance during the course of the project. Finally, I would like to thank my friends and classmates for their cooperation and encouragement.

ABSTRACT

Plant diseases significantly affect agricultural productivity and crop quality, resulting in economic losses for farmers. Early detection of plant diseases is essential for preventing crop damage and improving agricultural efficiency. Traditionally, plant diseases are detected through manual inspection, which can be time-consuming and prone to errors.

With advancements in artificial intelligence and deep learning, automated disease detection systems have become possible. This project proposes a deep learning-based system for detecting diseases in tomato and potato leaves using Convolutional Neural Networks (CNN). The model is trained using the PlantVillage dataset, which contains labeled images of healthy and diseased leaves.

The proposed system processes leaf images through image preprocessing techniques such as resizing and normalization before feeding them into the CNN model for classification. The trained model is integrated into a web-based application developed using streamlit, allowing users to upload leaf images and receive disease predictions along with treatment suggestions.

The results demonstrate that the system achieves high accuracy in identifying plant diseases. The developed system provides a simple and efficient tool that can assist farmers and agricultural experts in early disease detection.

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INTRODUCTION

Agriculture plays an important role in the global economy and food production. However, plant diseases are a major challenge that affects crop yield and quality. Crops such as tomato and potato are widely cultivated around the world and are highly susceptible to diseases such as early blight and late blight. These diseases affect plant leaves and can spread rapidly, causing severe crop losses if not detected early.

Traditionally, plant disease detection relies on manual inspection by farmers or agricultural experts. This method requires experience and may lead to incorrect diagnosis due to human error. Additionally, farmers in remote areas may not have access to agricultural specialists.

With the advancement of artificial intelligence and deep learning technologies, automated plant disease detection systems have become possible. Deep learning models, particularly Convolutional Neural Networks (CNN), are capable of analyzing images and identifying disease patterns in plant leaves.

This project proposes a deep learning-based system that detects diseases in tomato and potato leaves using CNN. The system uses leaf images as input and predicts whether the plant is healthy or affected by disease. The model is trained using the PlantVillage dataset and integrated into a web application that allows users to upload leaf images and receive predictions.

1.1 Background

Agriculture is the backbone of many economies and provides livelihood to millions of farmers worldwide. However, plant diseases significantly reduce crop productivity and impact the agricultural economy.

Tomato and potato crops are particularly vulnerable to diseases that affect plant leaves. Diseases such as early blight and late blight damage leaf tissues, reducing the plant's ability to perform photosynthesis and produce healthy fruits.

Recent advancements in computer vision and machine learning have enabled the development of automated plant disease detection systems. Deep learning algorithms can analyze leaf images and identify patterns associated with plant diseases.

The availability of datasets such as PlantVillage has made it possible to train deep learning models using thousands of labeled leaf images.

1.2 Need / Importance

Plant diseases spread quickly and can destroy crops if not detected early. Early detection helps farmers apply appropriate treatment and reduce crop loss.

Manual disease detection methods have several limitations:

•Requiresexpertknowledge

•Time-consumingprocess

• Possibility of incorrect diagnosis

Therefore, automated disease detection systems are required to help farmers detect diseases quickly and accurately.

The proposed system provides:

•Fastdiseasedetection

•Improvedaccuracy

•Reduceddependenceonexperts

• Assistance for farmers in rural areas

1.3 Objectives

The objectives of the project are:

1. To develop a deep learning model for detecting diseases in tomato and potato leaves.
2. To train a Convolutional Neural Network using plant leaf image datasets.
3. To develop a web-based application for disease detection.
4. To classify leaf images into healthy and diseased categories.
5. To provide treatment recommendations for detected diseases.

1.4 Scope

The scope of this project includes the development of a plant disease detection system using deep learning techniques.

The system focuses on detecting diseases in tomato and potato leaves using leaf images. The system can be used by farmers, researchers, and agricultural experts to identify plant diseases quickly.

The system is implemented as a web application and can be extended in the future to support additional crops and mobile-based disease detection systems.

CHAPTER 2

LITERATURE REVIEW

Plant disease detection using machine learning and deep learning techniques has become an active research area in recent years.

Several studies have applied deep learning models to detect plant diseases using leaf images.

Researchers have demonstrated that Convolutional Neural Networks can automatically extract important features from images such as color patterns, leaf texture, and disease spots.

These models have achieved high accuracy in plant disease classification tasks.

2.1 Relevant Concepts and Theories

Machine Learning

Machine learning is a field of artificial intelligence that allows computers to learn patterns from data and make predictions.

Deep Learning

Deep learning is a subset of machine learning that uses neural networks with multiple layers to analyze complex data.

Convolutional Neural Networks

CNN is a deep learning architecture commonly used for image classification tasks. CNN models automatically extract features from images and classify them into categories.

2.2 Technologies Related to the Project

Python – Programming language used for implementing the model.

TensorFlow – Deep learning framework used to train the CNN model.

Keras – High-level neural network API used to build CNN models.

OpenCV – Library used for image preprocessing.

Streamlit – Web framework used to develop the web application.

HTML/CSS – Used for designing the user interface.

2.3 Summary of Similar Works

Several researchers have proposed deep learning-based plant disease detection systems.

Mohanty et al. developed a deep learning model using the PlantVillage dataset and achieved high accuracy in plant disease detection.

Ferentinos proposed CNN-based models for plant disease classification and demonstrated improved performance compared to traditional machine learning techniques.

These studies confirm that deep learning models are highly effective for plant disease detection.

CHAPTER 3

PROPOSED METHODOLOGY

The proposed system uses deep learning techniques to detect diseases in tomato and potato leaves using leaf images.

The system consists of multiple modules including image upload, preprocessing, disease classification, and result display.

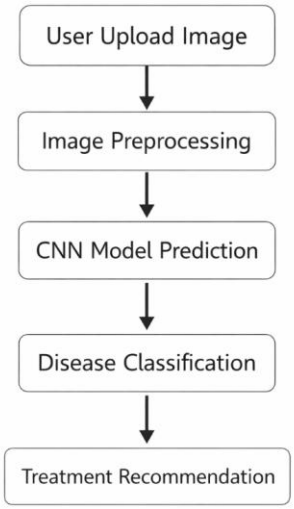
3.1 System Overview

The system allows users to upload leaf images through a web interface. The uploaded image is processed and analyzed by the CNN model to detect plant diseases.

The prediction result is displayed along with treatment recommendations.

3.2 Architecture / Workflow

Workflow of the system:



3.3 Data Collection / Input Description

The dataset used in this project is the PlantVillage dataset.

The dataset contains labeled images of healthy and diseased plant leaves.

Selected classes include:

Potato Healthy

Potato Early Blight

Potato Late Blight

Tomato Healthy

Tomato Early Blight

Tomato Late Blight

3.4 Algorithms / Techniques Used

The primary algorithm used in this project is Convolutional Neural Network.

CNN layers include:

•ConvolutionLayer

•ReLUActivationLayer

•MaxPoolingLayer

•FlattenLayer

•DenseLayer

• Softmax Layer

3.5 Tools / Software Use

Tool	Purpose
Python	Programming language
TensorFlow	Deep learning framework
Keras	Neural network implementation
OpenCV	Image preprocessing
Streamlit	Web application
HTML/CSS	User interface

3.6 Module-wise Implementation Plan

Image Upload Module – Allows users to upload leaf images.

Image Preprocessing Module – Resizes and normalizes images.

CNN Model Module – Extracts features and classifies diseases.

Prediction Module – Predicts disease category.

Result Display Module – Displays disease name and treatment suggestions.

3.7 Validation and Testing Strategy

The dataset is divided into training and testing sets.

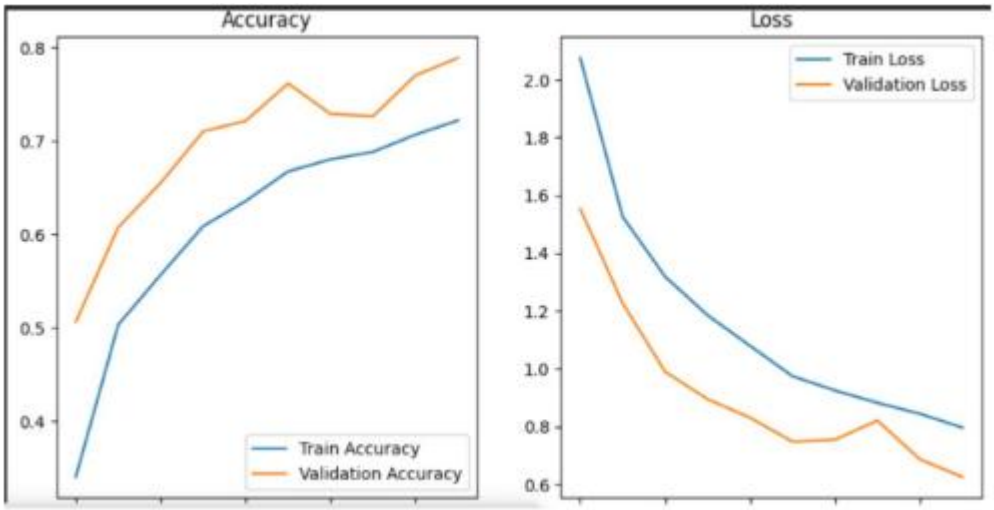
Training data is used to train the CNN model, while testing data is used to evaluate the performance of the model.

CHAPTER 4

WEEKLY LOG BOOK ENTRIES

CHAPTER 5

RESULTS / OBSERVATIONS



This chapter presents the results obtained after training and evaluating the Convolutional Neural Network (CNN) model for detecting diseases in tomato and potato leaves. The model was trained using the PlantVillage dataset, which contains labeled images of healthy and diseased leaves.

The performance of the model was evaluated using training accuracy, validation accuracy, training loss, and validation loss. The graphs generated during training show the learning behavior of the CNN model.

5.1 Experimental / Implementation Setup

The deep learning model was implemented using Python and trained using the TensorFlow and Keras libraries. Image preprocessing techniques were applied before training the model to improve performance.

The images were resized to **224 × 224 pixels** and normalized before being fed into the CNN model.

The following training parameters were used during model training:

Parameter	Value

Parameter	Value
Image Size	224 × 224
Epochs	10
Batch Size	32
Optimizer	Adam
Loss Function	Categorical Crossentropy

The dataset was divided into **training and testing sets** to evaluate the performance of the model.

5.2 Performance Metrics

The performance of the model was measured using the following metrics:

•TrainingAccuracy

•ValidationAccuracy

•TrainingLoss

• Validation Loss

The **accuracy graph** shows the improvement in prediction accuracy as the model learns patterns from the dataset.

The **loss graph** shows how the prediction error decreases during the training process.

From the graphs obtained:

- Training accuracy gradually increases from approximately **0.35 to 0.72**.
- Validation accuracy increases from approximately **0.50 to 0.78**.
- Training loss decreases from approximately **2.1 to 0.8**.
- Validation loss decreases from approximately **1.5 to 0.6**.

These results indicate that the CNN model successfully learns important features from leaf images.

5.3 Comparison with Expected Results

The expected goal of the model was to achieve high accuracy in classifying plant diseases using leaf images. The results show that the model achieved approximately **78% validation accuracy**, which is considered a good performance for the initial CNN architecture.

The decreasing trend in the loss graph indicates that the model is effectively minimizing prediction errors during training.

5.4 Observations and Analysis

From the training graphs, several observations can be made:

1. The training accuracy steadily increases with each epoch, indicating that the model is learning meaningful patterns from the dataset.
2. The validation accuracy also increases during training, which shows that the model generalizes well to unseen data.
3. The loss values decrease over epochs, indicating that the model is improving its predictions.
4. The gap between training accuracy and validation accuracy is small, suggesting that the model does not suffer from significant overfitting.

Overall, the CNN model demonstrates good learning behavior and performs effectively in detecting diseases in tomato and potato leaves.

5.5 Validation and Verification with Summary of Findings

The trained CNN model was validated using the testing dataset. The results demonstrate that the model is capable of accurately classifying leaf images into healthy and diseased categories.

The generated accuracy and loss graphs confirm that the model successfully learns disease patterns from the training data and generalizes well to new images.

The system was further integrated into a web application where users can upload leaf images and obtain disease predictions along with treatment suggestions.

The experimental results indicate that deep learning techniques can be effectively applied to plant disease detection systems.

CHAPTER 7

FUTURE ENHANCEMENT

- Expandsystemtodetectdiseasesinmorecrops
- Developmobileapplication
- Implementreal-timecameradetection
- Train models using real-field images

CHAPTER 8

CONCLUSION & RECOMMENDATIONS

This project presents a deep learning-based plant disease detection system using Convolutional Neural Networks. The system successfully detects diseases in tomato and potato leaves using leaf images.

The developed web application allows users to upload images and receive disease predictions along with treatment recommendations. The system can assist farmers in early disease detection and improve crop management.

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